DISSERTATION PROPOSAL

1. Proposed Title of the Dissertation

Co-Evolutionary Malware Detection Using Reinforcement Learning and Adaptive Defensive Retraining

2. Problem Domain / Area

Cybersecurity and Artificial Intelligence, specifically Adversarial Machine Learning applied to malware detection and endpoint security. The work sits at the intersection of reinforcement learning, adversarial robustness, and applied network/endpoint defense.

3. Motivation for Topic Selection (Objective)

Machine-learning-based malware classifiers have become the backbone of modern endpoint detection and response (EDR) products, but research since 2018 has repeatedly shown that reinforcement-learning (RL) agents can learn functionality-preserving file modifications that reliably evade these classifiers in fully black-box settings. This trend is highly relevant today as attackers increasingly automate evasion pipelines, turning adaptive, self-improving malware into a practical, escalating threat rather than a theoretical one.

The topic aligns closely with a specialization in AI-driven cybersecurity, drawing together reinforcement learning, adversarial machine learning, and applied security engineering into a single research problem. The personal motivation for this topic is a specific dissatisfaction with how the existing literature is framed: most work reports a one-time evasion success rate against a frozen classifier and stops there, leaving open the more practically important question of what happens when the defender is allowed to adapt too.

The potential research contribution lies in reframing malware detection as a continuous, iterated contest between an evolving attacker and a retraining defender, rather than a single-shot classification task. This has direct, practical implications for how vendors of ML-based detection products should schedule and structure model retraining.

4. Statement of Problem (Description)

Existing reinforcement-learning-based malware evasion frameworks (e.g., gym-malware, DQEAF, MERLIN) evaluate an attacker's ability to defeat a single, static, frozen classifier and treat the exercise as complete once evasion is demonstrated. It remains unclear whether a defender that periodically retrains on newly evolved evasive samples converges toward a stable, robust decision boundary, oscillates indefinitely against a continually adapting attacker, or is perpetually outpaced. This dissertation investigates malware detection as a continuous co-evolutionary game between an RL-based evasion agent and an adaptively retrained classifier, aiming to characterize the conditions under which stable, robust detection can be achieved.

5. Novelty / Uniqueness of the Topic

- Prior work (gym-malware, DQEAF, MERLIN, and related frameworks) is dominated by single-round, black-box evasion studies conducted against a static, frozen detector.
- Very little published work formalizes the attacker–defender relationship as a repeated or Stackelberg game with alternating retraining rounds, and even less studies the resulting convergence, oscillation, or divergence behavior across rounds.
- The proposed contribution is methodological rather than purely algorithmic: (a) a co-evolutionary training loop pairing an RL attacker with an incrementally retrained defender, (b) empirical characterization of the resulting game dynamics, and (c) practical guidance on retraining cadence and strategy for building detectors that remain robust to a moving adversary.
- Single-round evasion has been extensively studied since 2018; the iterated, co-evolutionary defense setting remains largely open in the literature, which is the specific gap this dissertation targets.

6. Mapping with Sustainable Development Goals (SDGs)

SDG Number & Title	Justification
SDG 9 – Industry, Innovation and Infrastructure	Advances resilient digital infrastructure by developing adaptive, self-updating security systems capable of withstanding evolving AI-driven cyber threats.
SDG 16 – Peace, Justice and Strong Institutions	Strengthens institutional resilience against cybercrime by improving the robustness of malware defenses that protect financial, governmental, and public digital infrastructure.

8. Expected Applications / Use Cases

- Hardening commercial EDR and antivirus engines against continuously evolving, RL-driven evasion techniques.
- Informing retraining and update-cadence policy for vendors of ML-based malware detection products.
- Strengthening malware defense pipelines in financial services and cloud workload protection platforms (CWPP).
- Supporting critical-infrastructure-adjacent IT environments that require detectors robust to sustained, adaptive adversarial pressure.

9. Pre-Requisite Skills

- Python programming with PyTorch/TensorFlow and scikit-learn
- Reinforcement learning fundamentals (DQN, policy gradient/REINFORCE, PPO)
- Static malware analysis and Portable Executable (PE) file format familiarity
- Game theory fundamentals (repeated games, Stackelberg/Nash equilibria)
- Experience with sandboxed/VM-based experimentation for security research
- Applied statistics and experimental evaluation methodology

10. Outcome

- A working co-evolutionary attacker–defender simulation framework, extending existing open-source RL evasion environments.
- Empirical characterization of convergence, oscillation, or divergence behavior of detectors under iterated retraining.
- A proposed retraining-cadence guideline for building malware classifiers robust to adaptive adversaries.
- A dissertation manuscript and at least one paper-length contribution suitable for submission to an adversarial ML / security venue.

11. Preliminary Keywords

Adversarial Machine Learning; Reinforcement Learning; Malware Evasion; Co-evolutionary Game Theory; Endpoint Detection and Response (EDR); Robust Classification

12. References / Sources Checked

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